

Penetration Testing Project: VULNER

Program Code: ZX301

Student Name: David Rozi | **Student Code:** s16 | **Unit:** PERES25B | **Trainer:** Tzah Azoalis

Project Title:

Automated Network Vulnerability Scanner

Submission Files:

PERES25B.s16.zx301.sh

PERES25B.s16.zx301.pdf

1. Project Overview

This project implements an automated penetration testing support tool written in Bash. The purpose of the script is to scan a local authorized network, identify live hosts, detect open TCP and UDP services, check for weak credentials on common login services, and map potential vulnerabilities.

The script supports two modes:

Basic mode:

Performs host discovery, TCP service scan, UDP service scan, service version detection, and weak credential checks.

Full mode:

Includes everything in Basic mode, and also performs vulnerability mapping using Nmap Scripting Engine and Searchsploit.

The project was developed for an authorized training environment only. The tool is intended for defensive security testing, lab validation, and controlled penetration testing exercises.

2. Environment Preparation

The project environment was prepared on Kali Linux.

A dedicated working directory was created under:

/home/david/zx301-vulner-project

Inside this directory, the following project structure was created:

docs/

results/

screenshots/

vulner.sh

password.lst

The password.lst file contains a small built-in password list for weak credential testing in the authorized lab environment.

The vulner.sh file is the main Bash script that will be developed during this project.

The results directory will store all scan outputs.

The screenshots directory will store evidence screenshots for the final PDF report.

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The docs directory will store documentation-related files during development.

The required tools were also verified on Kali Linux. The tools nmap, hydra, searchsploit, zip, grep, awk, sed, and tee were found installed and ready for use.

```

david@Kali: ~/zx301-vulner-project
Session Actions Edit View Help
=====
ZX301 VULNER Project - Environment Preparation
=====

Student Name : David Rozi
Student Code : s16
Unit        : PERES25B
Trainer     : Tzah Azoalis
Program Code : ZX301

Working Directory:
/home/david/zx301-vulner-project

Project Structure:
.
./docs
./results
./screenshots

Main Project Files:
-rw-rw-r-- 1 david david 90 Jun  8 16:25 password.lst
-rwxrwxr-x 1 david david  0 Jun  8 16:25 vulner.sh

Required Tools Verification:
[OK] nmap found
[OK] hydra found
[OK] searchsploit found
[OK] zip found
[OK] grep found
[OK] awk found
[OK] sed found
[OK] tee found

Environment preparation completed successfully.

(david@Kali)-[~/zx301-vulner-project]
$
```

Figure 1: Clean environment preparation output showing the project directory, main files, and required tools verification.

3. User and Password Lists

The project uses two local text files for weak credential testing:

user.lst

password.lst

The user.lst file contains common usernames such as admin, administrator, root, guest, and service.

The password.lst file contains a small built-in password list with common weak passwords. This list is used only inside the authorized lab environment in order to test whether login services such as SSH, FTP, TELNET, and RDP allow weak credentials.

Using a built-in password list helps the script meet the project requirement of checking for weak passwords automatically. The script will also support a custom password list if the user chooses to provide one during execution.

This stage prepares the credential input files before developing the main Bash automation script.

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```

david@Kali: ~/zx301-vulner-project
Session Actions Edit View Help
=====
ZX301 VULNER Project - User and Password Lists
=====

Working Directory:
/home/david/zx301-vulner-project

User List:
admin
administrator
root
kali
user
test
guest
ftp
service
scanner

Password List:
admin
password
123456
12345678
qwerty
letmein
toor
root
kali
user
test
Passw0rd
Password1

List Files:
-rw-rw-r-- 1 david david 90 Jun  8 16:25 password.lst
-rw-rw-r-- 1 david david 66 Jun  8 16:33 user.lst

User and password lists created successfully.
=====

(david@Kali)-[~/zx301-vulner-project]
$
```

Figure 2: Built-in user and password lists prepared for weak credential testing.

4. Script Development

The main Bash script was created as `vulner.sh`.

The script starts with a clear project header that includes the project name, program code, student name, student code, unit, trainer name, and class code.

The script was developed using functions in order to keep the code organized, readable, and maintainable.

The main functions in the script are:

`check_tools`

`get_user_input`

`run_host_discovery`

`run_tcp_scan`

`run_udp_scan`

`run_weak_credentials`

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run_nse_vulnerability_scan

run_searchsploit_analysis

create_summary

search_inside_results

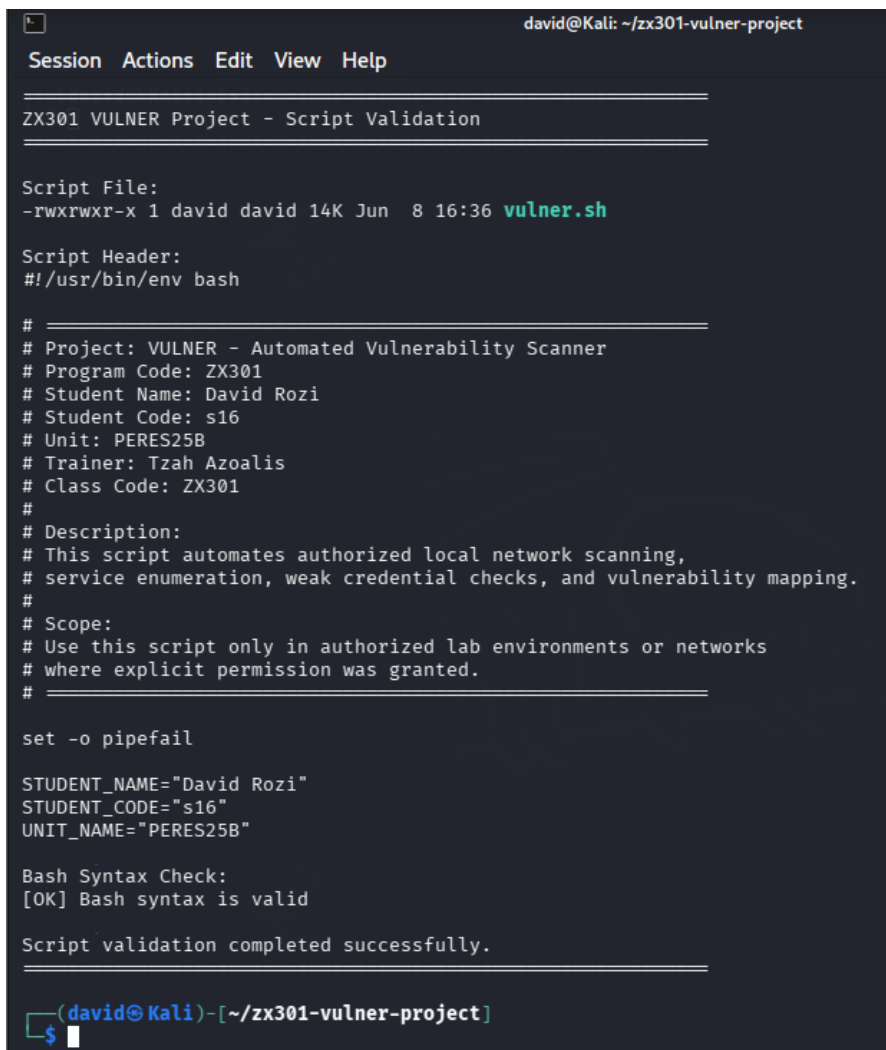
zip_results

The script validates the network input, checks that all required tools are installed, performs automated scans, saves results into a dedicated output directory, allows searching inside the results, and creates a ZIP archive with all generated scan files.

Before execution, the script was tested with Bash syntax validation using:

```
bash -n vulner.sh
```

The validation completed successfully, which confirms that the script does not contain Bash syntax errors.



```
david@Kali: ~/zx301-vulner-project
Session Actions Edit View Help
=====
ZX301 VULNER Project - Script Validation
=====

Script File:
-rwxrwxr-x 1 david david 14K Jun  8 16:36 vulner.sh

Script Header:
#!/usr/bin/env bash

# =====
# Project: VULNER - Automated Vulnerability Scanner
# Program Code: ZX301
# Student Name: David Rozi
# Student Code: s16
# Unit: PERES25B
# Trainer: Tzah Azoalis
# Class Code: ZX301
#
# Description:
# This script automates authorized local network scanning,
# service enumeration, weak credential checks, and vulnerability mapping.
#
# Scope:
# Use this script only in authorized lab environments or networks
# where explicit permission was granted.
# =====

set -o pipefail

STUDENT_NAME="David Rozi"
STUDENT_CODE="s16"
UNIT_NAME="PERES25B"

Bash Syntax Check:
[OK] Bash syntax is valid

Script validation completed successfully.
=====

(david@Kali)-[~/zx301-vulner-project]
$
```

Figure 3: Bash syntax validation completed successfully for the main vulner.sh script.

Penetration Testing Project: VULNER

```
ZX301 VULNER Project - Script Functions Overview

Script:
-rwxrwxr-x 1 david david 14K Jun  8 16:36 vulner.sh

Functions implemented in vulner.sh:
48:print_banner() {
61:print_stage() {
66:print_info() {
70:print_warn() {
74:die() {
79:check_tools() {
99:validate_network() {
112:require_private_network() {
129:normalize_mode() {
146:sanitize_name() {
150:get_user_input() {
212:run_privileged_nmap() {
220:run_host_discovery() {
240:run_tcp_scan() {
246:run_udp_scan() {
254:run_weak_credentials() {
347:run_nse_vulnerability_scan() {
353:run_searchsploit_analysis() {
366:create_summary() {
423:search_inside_results() {
450:zip_results() {
462:fix_permissions() {
472:main() {

Function overview completed successfully.
```

Figure 4: Main functions implemented in the vulner.sh Bash script.

The script was divided into multiple Bash functions in order to keep the code organized, readable, and easier to maintain.

Each function is responsible for a specific part of the automation process, such as tool verification, user input validation, host discovery, TCP and UDP scanning, weak credential testing, vulnerability scanning, result summarization, result searching, and ZIP archive creation.

This structure helps prove that the script was not written as one long block of commands, but as a structured automation tool.

5. Network Range Identification

Before running the scanner, the local network range was identified from Kali Linux.

The command output showed that the Kali machine is using the eth0 interface with the IP address:

192.168.47.136/24

The routing table showed the local network route:

192.168.47.0/24

Therefore, the authorized lab network selected for the scan is:

192.168.47.0/24

This step is important because the script requires the user to provide a valid network range before scanning. Using the correct CIDR range prevents scanning the wrong network and keeps the test within the authorized lab scope.

Penetration Testing Project: VULNER

```

Session Actions Edit View Help
=====
ZX301 VULNER Project - Network Range Identification
=====

IPv4 Addresses:
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    inet 192.168.47.136/24 brd 192.168.47.255 scope global dynamic noprefixroute eth0
        valid_lft 1150sec preferred_lft 1150sec

Routing Table:
default via 192.168.47.2 dev eth0 proto dhcp src 192.168.47.136 metric 100
192.168.47.0/24 dev eth0 proto kernel scope link src 192.168.47.136 metric 100

Network identification completed.
=====
```

Figure 5: Network range identification showing Kali Linux IP address and the authorized lab CIDR range.

6. Basic Scan Execution

The script was executed in Basic mode against the authorized lab network:

192.168.47.0/24

The Basic scan performed the following automated actions:

1. Verified that all required tools were installed.
2. Collected the network range, output directory name, scan mode, and password list choice from the user.
3. Performed host discovery using Nmap.
4. Performed TCP service scanning with service version detection.
5. Performed UDP service scanning on the top UDP ports.
6. Checked weak credentials on supported login services using Hydra.
7. Created a final summary file.
8. Created a ZIP archive containing the scan results.

The scan discovered 4 live hosts in the lab network:

192.168.47.1
192.168.47.136
192.168.47.2
192.168.47.254

The TCP scan identified the following open services on 192.168.47.136:

FTP on port 21 running vsftpd 3.0.5
SSH on port 22 running OpenSSH 10.2p1 Debian 3
HTTP on port 80 running Apache httpd 2.4.66

The weak credential test identified one weak FTP credential:

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Username: ftp
Password: admin

This finding demonstrates that the script successfully performed weak credential testing as required by the project instructions.

At the end of the scan, the script generated a results directory and compressed all result files into a ZIP archive.

```
=====
ZX301 VULNER Project - Basic Scan Evidence
=====

Results Directory:
results/basic_scan_basic_20260608_164908

Scan Configuration:
Network      : 192.168.47.0/24
Scan Mode    : Basic
Password List: /home/david/zx301-vulner-project/password.lst
User List     : /home/david/zx301-vulner-project/user.lst

Live Hosts Found:
 1 192.168.47.1
 2 192.168.47.136
 3 192.168.47.2
 4 192.168.47.254

Detected TCP Services:
192.168.47.136:21/tcp  ftp  vsftpd 3.0.5
192.168.47.136:22/tcp  ssh  OpenSSH 10.2p1 Debian 3 (protocol 2.0)
192.168.47.136:80/tcp  http  Apache httpd 2.4.66 ((Debian))

Weak Credential Finding:
Host: 192.168.47.136 | Service: FTP | Username: ftp | Password: admin

ZIP Archive:
-rw-rw-r-- 1 david david 14K Jun  8 17:00 results/basic_scan_basic_20260608_164908.zip

Basic scan completed successfully.
=====
```

Figure 6: Basic scan evidence showing scan configuration, live hosts, detected TCP services, weak credential finding, and ZIP archive creation.

6.1 Generated Basic Scan Result Files

After the Basic scan completed successfully, the script generated multiple result files inside the output directory.

The generated files include Nmap output formats, Hydra result files, a live hosts file, a run log, and a final summary file.

The script also created a ZIP archive that contains the Basic scan results.

This proves that the script saves the scan evidence automatically and organizes the results in a structured output directory.

```
Session  Actions  Edit  View  Help

ZX301 VULNER Project - Generated Result Files

Results Directory:
results/basic_scan_basic_20260608_164908

Generated Files:
results/basic_scan_basic_20260608_164908/01_host_discovery.gnmap
results/basic_scan_basic_20260608_164908/01_host_discovery.nmap
results/basic_scan_basic_20260608_164908/01_host_discovery.xml
results/basic_scan_basic_20260608_164908/02_tcp_services.gnmap
results/basic_scan_basic_20260608_164908/02_tcp_services.nmap
results/basic_scan_basic_20260608_164908/02_tcp_services.xml
results/basic_scan_basic_20260608_164908/03_udp_services.gnmap
results/basic_scan_basic_20260608_164908/03_udp_services.nmap
results/basic_scan_basic_20260608_164908/03_udp_services.xml
results/basic_scan_basic_20260608_164908/hydra_results/192.168.47.136_21_ftp.txt
results/basic_scan_basic_20260608_164908/hydra_results/192.168.47.136_22_ssh.txt
results/basic_scan_basic_20260608_164908/live_hosts.txt
results/basic_scan_basic_20260608_164908/run.log
results/basic_scan_basic_20260608_164908/summary.md

ZIP File:
-rw-rw-r-- 1 david david 14K Jun  8 17:00 results/basic_scan_basic_20260608_164908.zip

Generated result files verification completed successfully.
```

Figure 7: Generated Basic scan result files and ZIP archive verification.

7. Full Scan Execution

The script was executed again in Full mode against the authorized lab network:

192.168.47.0/24

Full mode includes all Basic mode actions and adds vulnerability mapping using Nmap Scripting Engine and Searchsploit.

During the Full scan, the script performed the following actions:

1. Verified the required tools.
2. Collected user input.
3. Performed host discovery.
4. Performed TCP service scanning with version detection.
5. Performed UDP service scanning.
6. Checked weak credentials on supported login services.
7. Executed Nmap NSE vulnerability scripts.
8. Executed Searchsploit analysis based on the Nmap XML scan results.
9. Created a final summary.
10. Created a ZIP archive with all result files.

The Full scan confirmed the same live hosts and services identified in the Basic scan.

The script also identified the weak FTP credential:

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Username: ftp
Password: admin

In addition, Full mode created dedicated output files for NSE vulnerability scanning and Searchsploit analysis. This confirms that the Full mode executed the required vulnerability mapping stage.

```
Session  Actions  Edit  View  Help
=====
ZX301 VULNER Project - Full Scan Evidence
=====

Results Directory:
results/full_scan_full_20260608_172436

Scan Configuration:
Network      : 192.168.47.0/24
Scan Mode    : Full
Password List: /home/david/zx301-vulner-project/password.lst
User List    : /home/david/zx301-vulner-project/user.lst

Live Hosts Found:
1 192.168.47.1
2 192.168.47.136
3 192.168.47.2
4 192.168.47.254

Detected TCP Services:
192.168.47.136:21/tcp  ftp  vsftpd 3.0.5
192.168.47.136:22/tcp  ssh  OpenSSH 10.2p1 Debian 3 (protocol 2.0)
192.168.47.136:80/tcp  http  Apache httpd 2.4.66 ((Debian))

Weak Credential Finding:
Host: 192.168.47.136 | Service: FTP | Username: ftp | Password: admin

Full Mode Checks:
[OK] NSE vulnerability scan output exists
[OK] Searchsploit analysis output exists

Searchsploit Preview:
Exploit Title                                     | Path
vsftpd 2.0.5 - 'CWD' (Authenticated) Remote Memory Consumption | linux/dos/5814.pl
vsftpd 2.0.5 - 'deny_file' Option Remote Denial of Service (1) | windows/dos/31818.sh
vsftpd 2.0.5 - 'deny_file' Option Remote Denial of Service (2) | windows/dos/31819.pl
vsftpd 2.3.2 - Denial of Service                          | linux/dos/16270.c
vsftpd 2.3.4 - Backdoor Command Execution                 | unix/remote/49757.py
vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)      | unix/remote/17491.rb
vsftpd 3.0.3 - Remote Denial of Service                    | multiple/remote/49719.py
Shellcodes: No Results
Exploit Title                                     | Path
Debian OpenSSH - (Authenticated) Remote SELinux Privilege Escalation | linux/remote/6094.txt
Dropbear / OpenSSH Server - 'MAX_UNAUTH_CLIENTS' Denial of Service   | multiple/dos/1572.pl

ZIP Archive:
-rw-rw-r-- 1 david david 24K Jun  8 17:38 results/full_scan_full_20260608_172436.zip

Full scan completed successfully.

(david@Kali) - [~/zx301-vulner-project]
$
```

Figure 8: Full scan evidence showing scan configuration, live hosts, TCP services, weak FTP credential, NSE output, Searchsploit output, and ZIP archive creation.

8. Search Inside Results

The script includes an option that allows the user to search inside the generated result files.

This feature helps the user quickly locate specific services, findings, vulnerabilities, or credentials inside the scan outputs.

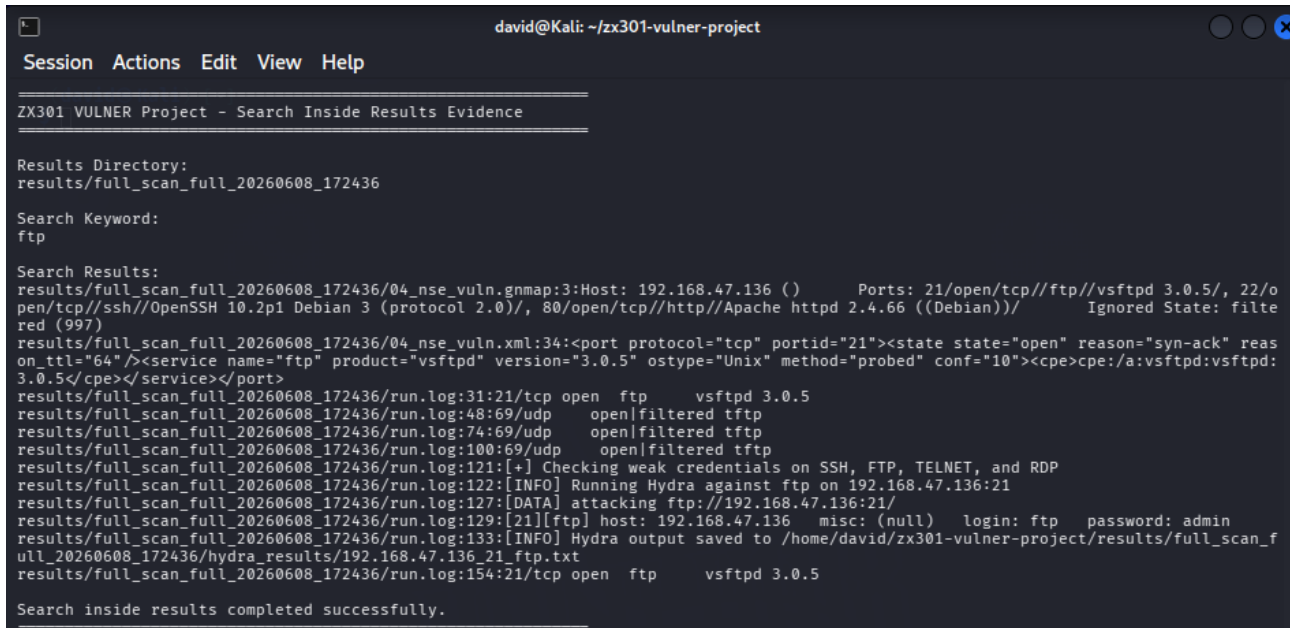
To demonstrate this functionality, the generated Full scan result directory was searched for the keyword:

ftp

Penetration Testing Project: VULNER

The search returned relevant results from the generated files, including FTP service information and weak credential evidence.

This confirms that the script meets the project requirement of allowing the user to search inside the results.



```
david@kali: ~/zx301-vulner-project
Session Actions Edit View Help

ZX301 VULNER Project - Search Inside Results Evidence

Results Directory:
results/full_scan_full_20260608_172436

Search Keyword:
ftp

Search Results:
results/full_scan_full_20260608_172436/04_nse_vuln.gnmap:3:Host: 192.168.47.136 () Ports: 21/open/tcp//ftp//vsftpd 3.0.5/, 22/o
pen/tcp//ssh//OpenSSH 10.2p1 Debian 3 (protocol 2.0)/, 80/open/tcp//http//Apache httpd 2.4.66 ((Debian))/ Ignored State: filte
red (997)
results/full_scan_full_20260608_172436/04_nse_vuln.xml:34:<port protocol="tcp" portid="21"><state state="open" reason="syn-ack" reas
on_ttl="64"/><service name="ftp" product="vsftpd" version="3.0.5" ostype="Unix" method="probed" conf="10"><cpe>cpe:/a:vsftpd:vsftpd:
3.0.5</cpe></service></port>
results/full_scan_full_20260608_172436/run.log:31:21/tcp open ftp vsftpd 3.0.5
results/full_scan_full_20260608_172436/run.log:48:69/udp open|filtered tftp
results/full_scan_full_20260608_172436/run.log:74:69/udp open|filtered tftp
results/full_scan_full_20260608_172436/run.log:100:69/udp open|filtered tftp
results/full_scan_full_20260608_172436/run.log:121:[+] Checking weak credentials on SSH, FTP, TELNET, and RDP
results/full_scan_full_20260608_172436/run.log:122:[INFO] Running Hydra against ftp on 192.168.47.136:21
results/full_scan_full_20260608_172436/run.log:127:[DATA] attacking ftp://192.168.47.136:21/
results/full_scan_full_20260608_172436/run.log:129:[21][ftp] host: 192.168.47.136 misc: (null) login: ftp password: admin
results/full_scan_full_20260608_172436/run.log:133:[INFO] Hydra output saved to /home/david/zx301-vulner-project/results/full_scan_f
ull_20260608_172436/hydra_results/192.168.47.136_21_ftp.txt
results/full_scan_full_20260608_172436/run.log:154:21/tcp open ftp vsftpd 3.0.5

Search inside results completed successfully.
```

Figure 9: Search inside results evidence showing keyword search across generated scan files.

9. Final Submission Files

According to the project submission requirements, the final submission must include two files only:

PERES25B.s16.zx301.sh
PERES25B.s16.zx301.pdf

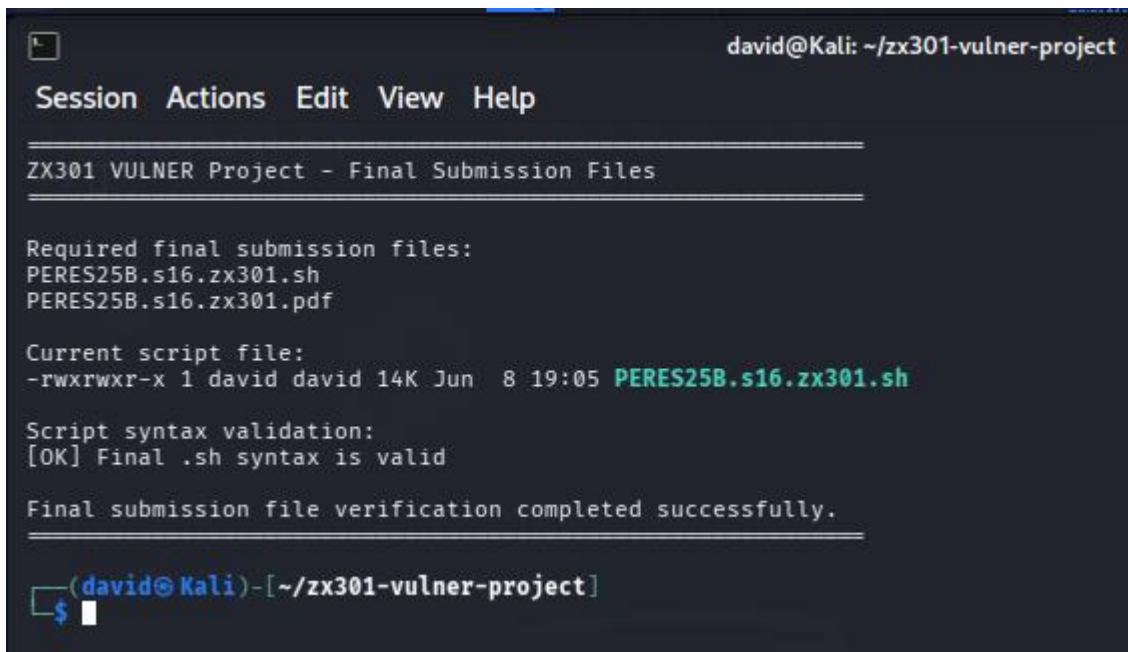
The Bash script was copied to the required final filename and tested again using Bash syntax validation.

The syntax validation completed successfully, confirming that the final .sh file is ready for submission.

The current DOCX file will be exported to PDF and submitted as:

PERES25B.s16.zx301.pdf

Penetration Testing Project: VULNER



```

david@Kali: ~/zx301-vulner-project
Session Actions Edit View Help
=====
ZX301 VULNER Project - Final Submission Files
=====

Required final submission files:
PERES25B.s16.zx301.sh
PERES25B.s16.zx301.pdf

Current script file:
-rwxrwxr-x 1 david david 14K Jun  8 19:05 PERES25B.s16.zx301.sh

Script syntax validation:
[OK] Final .sh syntax is valid

Final submission file verification completed successfully.
=====

(david@Kali)-[~/zx301-vulner-project]
$
```

Figure 10: Final script file verification using the required submission filename.

10. Conclusion

The VULNER project successfully implements an automated Bash-based penetration testing support tool.

The script receives user input, validates the network range, supports both Basic and Full scan modes, performs host discovery, scans TCP and UDP services, detects service versions, checks weak credentials, maps potential vulnerabilities using NSE and Searchsploit, allows result searching, and creates a ZIP archive containing all generated results.

The project was tested in an authorized local lab network only.

The screenshots and generated output files prove that the script works as expected and meets the main project requirements.